Electromagnetism

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Electromagnetism (A29041)

Short Title:	Electromagnetism
Faculty	Science and Computing
Department	Science
Cluster	Mathematics and Physics
Author	MALHOURANI
Credits	5

Level: Introductory

Description of Module / Aims

This module develops the student's understanding of electromagnetism.

Programmes

ELEC-0063	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	2 / 4 / M

Indicative Content

- Electric force, field and potential, Gauss's law, capacitors and dielectrics.
- EMF, DC circuits, Kirchoff's laws and Wheatstone bridge;
- Magnetic fields, force on a moving charge, force on a current carrying wire. Biot-Savart and Ampere's laws. Hall Effect.
- Electric and magnetic dipoles.
- Electromagnetic induction, Faraday's and Lenz's Laws.
- Transformer and basic AC circuits
- Vector Calculus: Differentiation, integration, Line, surface and volume integrals, Div, Grad, Curl.
- Divergence theorem, Stokes' Theorem.
- Maxwell equations integral and differential forms and the wave equation

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Apply electric and magnetic field laws to solve a range of problems.
2. Define and explain a range of basic terms in electromagnetism.
3. Use vector calculus to solve problems in electromagnetism.
4. Apply problem-solving and investigative techniques in a laboratory context.
5. Communicate practical work effectively in the form of laboratory reports.

Learning and Teaching Methods

- Lectures
- Tutorials
- laboratory work. The integrated practical programme is designed to run in parallel with lectures and allows the student to develop a range of experimental skills. The importance of planning of experiments, data acquisition, and data analysis will be emphasized throughout.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3
Continuous Assessment	30%	4,5

Assessment Criteria

Below 40%: Student has demonstrated a very limited knowledge of the course material. He/She is unable to carry out critical thinking of subject area.

40-49%: Student has demonstrated an adequate knowledge of the course material He/She is able to carry out some critical thinking of subject area.

50 -59%: shows a reasonable level of understanding of the various principles involved.

60 -69%: shows good knowledge of the area. Shows ability to reason and analyse problems in this area.

70%+: excellent knowledge of the subject area. Excellent reasoning skills and focused well thought out arguments. General Syllabus Information Module Code TBA Last Revised 17/5/10 Allocated Time (hours/week)

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	6	
Practical	18	
Independent Learning	87	

Essential Material

- Holliday, D., R. Resnick and J. Walker. _Fundamental of Physics._ . 10th. USA: Wiley, 2014.

Supplementary Material

- Folan, L.M. _Modern Physics and Technology for undergraduates_. USA: World Scientific, 2003.
- Mansfield, M. and C. O'Sullivan. _Understanding Physics_. 2nd. USA: Wiley, .
- Schey, H.M. _Div,Grad, Curl and all that_. 4th. USA: W.W. Norton, 2005.
- Shankar, R. _Fundamentals of Physics II - Electromagnetism, Optics and Quantum Mechanics._ . USA: Yale University Press, 2016.

Requested Resources

- SCIENCE LAB: Physics